

- 5 (a) The mammalian circulatory system is described as a closed double circulation.

Explain why it is called a *closed* **and** *double* circulation.

closed

.....

double

..... [2]

- (b) Fig. 5.1 shows a drawing of an external view of a mammalian heart.

Two cross-sections were made of the heart:

- section 1 was made across the line **A–B**.
- section 2 was made across the line **C–D**.

Drawings of the two sections were viewed from above as shown by the arrow on Fig. 5.1. Fig. 5.2 is a drawing of section **A–B**. Fig. 5.3 is a drawing of section **C–D**.

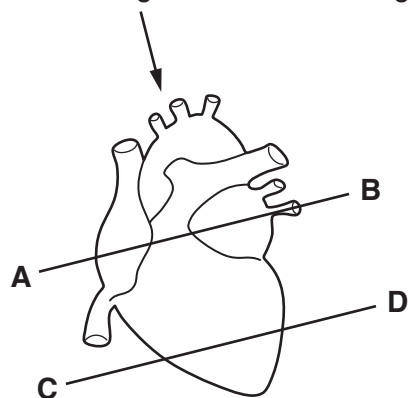


Fig. 5.1

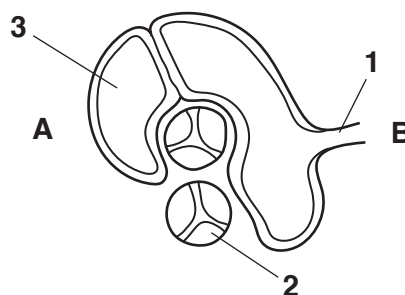


Fig. 5.2



Fig. 5.3

(i) Name structures **1**, **2** and **3**, as shown in Fig. 5.2.

1

2

3

[3]

(ii) Explain why the wall of chamber **Y** is thicker than the wall of chamber **X**, as shown in Fig. 5.3.

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..... [3]

(c) Explain how the contractions of the chambers of the heart are coordinated during one cardiac cycle.

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..... [4]

[Total: 12]